Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

SPECIAL PATIENT GROUPS

YG CG 601 MAY 2021

Pacemakers & Implantable Cardiac Defibrillators

REVIEW BY MAY 2025.

Author: Dr Stephan Clements
Reviewer: Cardiac Physiology Department, Departmental Guideline Group

Key Points

- **Cardiac Physiologist: 4058**
- **Pre-operative**
  - Identify pacemaker type & indication
  - Review ECG and CXR
  - Correct electrolyte disturbance
- **Intra-operative**
  - Magnets are located in theatre recovery
  - Defibrillator pads are preferably placed in A-P position
    - Anterior (negative) electrode: left sternal margin
    - Posterior (positive) electrode: inferior to left scapula

Within Normal Working Hours

- Contact the Cardiac Physiologist on 4058 for advice.
- Antitachycardia function should be disabled. It is unnecessary to deactivate rate-responsive modes unless Lithotripsy is planned.

Emergency management / out of hours

- **ICD's and Pacemakers with ICD function**
  - Have external defib available in the rare event that the ICD does not function when the magnet is removed.
  - Deactivate all ICD's where bipolar or monopolar is to be used by placing a magnet over it SECURELY.
  - In event of a shockable rhythm, reactivate the ICD by simply removing the magnet.
  - Note: With ICD's **that have** pacemaker function – the magnet will have NO EFFECT on its pacemaking function.

- ***Pacemakers (without ICD function)***
  - All modern pacemakers will go into asynchronous mode (fixed rate pacing AOO, VOO or DOO at a rate of 90 to 100 per minute) when a magnet is placed.
  - **Is the patient pacemaker dependent?**
    - This is difficult to determine from a single ECG, but if connected to a monitor you may be able to tell.
    - If you are not sure, it is safer to assume that the patient is pacemaker dependent.

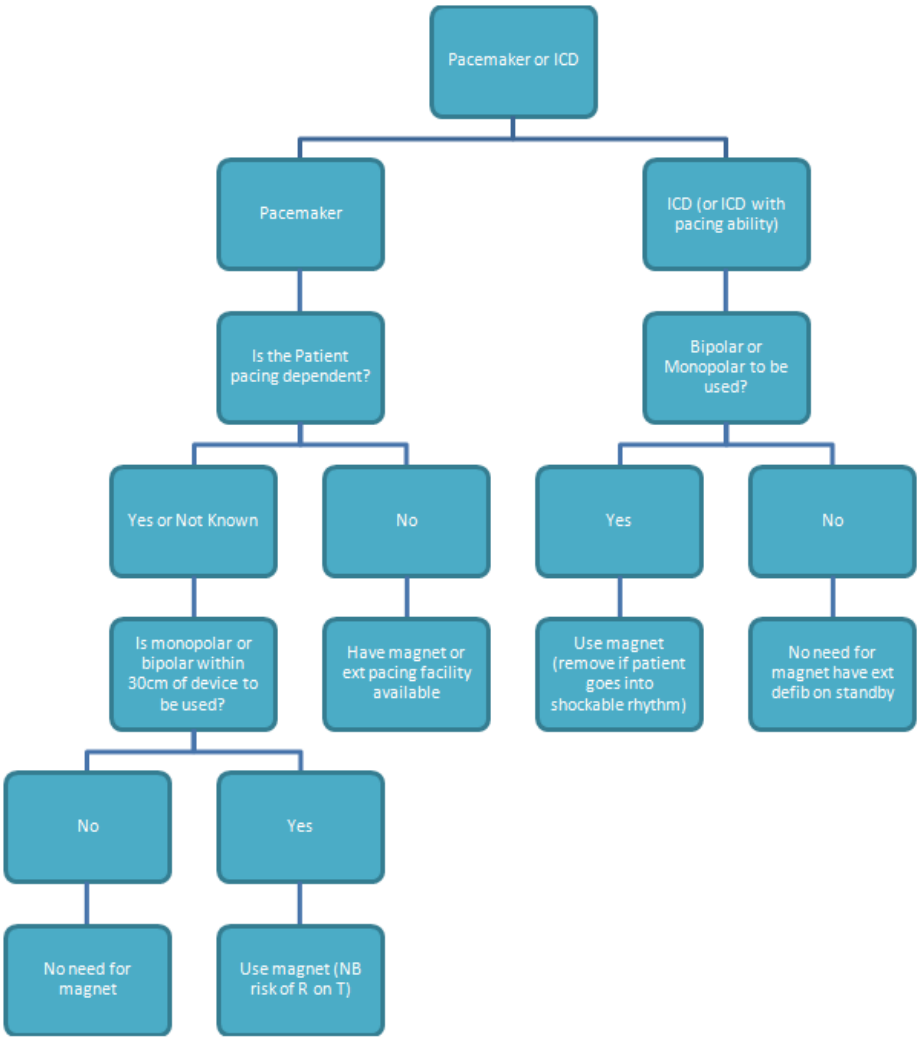
- **If dependant, is bipolar going to be within 30cm of the pacemaker or is monopolar going to be used?**
  - – If YES – PLACE A MAGNET.

NB. There is a theoretical risk of 'R on T' phenomenon when using fixed rate pacing, therefore have external defib pads available.

Follow Up

If no issues during the case then the patient does not need to be seen by the cardiac physiologists before their next routine check-up.

Summary Flow Chart



References

Bourke ME. The patient with a pacemaker or related device Can J Anaesth 1996,43;5:R24-R32